Carnegie Mellon University

Daksh Gupta <dakshg@andrew.cmu.edu>

Weekly Update - Vanguard / CMU Capstone Project

9 messages

Daksh Gupta <dakshg@andrew.cmu.edu> Mon, Sep 14, 2026 at 8:00 AM
To: "Song,Qiang [VCM]" <qiang_song@vcm.vanguard.com>, "Deshpande,Amit [VCM]" <amit_deshpande2@vcm.vanguard.com>
Cc: Anna Huang <annahua2@andrew.cmu.edu>, Caroline Liu <cmliu@andrew.cmu.edu>, Ziyu He <ziyuhe@andrew.cmu.edu>, Zola Zhang <zolaz@andrew.cmu.edu>, Chad Schafer <cschafer@cmu.edu>

Hi Amit and Qiang,

Following our discussion, we sat down as a team and worked through the research problem in more detail, focusing first on why the problem matters, what exactly we want to predict and understand, and then how we could approach it.

We have attached our working research plan, which lays out the problem statement, key research questions, proposed model design, and possible ways to evaluate and interpret the results. This is still a draft, and we would appreciate your feedback before we develop the methodology further.

Best,
Daksh (*On behalf of Team 6*)

On Wed, Sep 9, 2026 at 1:21 PM Daksh Gupta <dakshg@andrew.cmu.edu> wrote:

> Hi Qiang and Amit - Starting this thread to share weekly updates.
>
> Over the past week, our team reviewed the literature on bond fund performance and attribution. Our main learnings were:
>
> - Performance appears to persist, particularly among underperforming funds, but this does not necessarily indicate manager skill.
> - Existing evidence finds limited security-selection skills on average. Persistent outcomes may instead reflect duration, yield-curve, credit, liquidity, sector, and carry exposures, along with fees.
> - Evaluating skill therefore requires separating systematic exposures, timing or positioning, and security selection rather than focusing only on fund returns.
> - Holdings-based research in fixed income remains relatively limited, particularly at the security and benchmark-relative active-position level.
>
> Two papers were especially relevant to shaping our direction:
>
> - *Performance Evaluation with Portfolio Holdings Information* (Wermers, 2006) provides the conceptual foundation for using holdings to separate systematic exposures, timing, and security selection, and for testing persistence at the component level.
> - The recent working paper *Detecting Skilled Bond Fund Managers* applies a similar idea directly to bond funds using performance, fund characteristics, holdings, and machine learning. Since it closely aligns with our research question but is more focused on prediction but we could alter it towards attribution that is our focus here
>
> We can build on this framework by examining whether more granular metrics derived from N-PORT holdings, such as benchmark-relative active positions, duration, curve, credit, and sector exposures, can better explain the sources of persistent underperformance.
>
> Looking forward to speaking today.
>
> Best,
> Daksh

> On behalf of Team 6

Vanguard_CMU_Bond_Fund_Research_Plan.docx
36K

Daksh Gupta <dakshg@andrew.cmu.edu> Wed, Sep 23, 2026 at 8:00 AM
To: "Song,Qiang [VCM]" <qiang_song@vcm.vanguard.com>, "Deshpande,Amit [VCM]" <amit_deshpande2@vcm.vanguard.com>
Cc: Anna Huang <annahua2@andrew.cmu.edu>, Caroline Liu <cmliu@andrew.cmu.edu>, Ziyu He <ziyuhe@andrew.cmu.edu>, Zola Zhang <zolaz@andrew.cmu.edu>, Chad Schafer <cschafer@cmu.edu>

Hi Amit and Qiang,

Following our discussion about building monthly holdings snapshots, we have been looking into the SEC data and wanted to flag some limitations around reporting frequency, historical coverage, and consistency.

Late 2019 onwards: N-PORT provides more structured, machine-readable holdings data. However, the historical public filings generally cover only the final month of each fund's fiscal quarter, rather than a complete monthly holdings history. There is also a disclosure lag: March-end holdings, for example, may only become available in May. We would need to account for this timing to avoid using information before it was publicly available.

2004–2019: Quarterly holdings are generally available through a combination of Form N-Q and annual/semiannual shareholder reports, rather than N-PORT. Accessing these would require a separate extraction process. Compared with N-PORT, the main challenges are inconsistent document and table formats, less standardized security identifiers and classifications, and differences in the availability of bond characteristics and risk measures. We would also need to reconcile fund identities across filings and reporting regimes.

2000–2004: Holdings disclosures were generally semiannual. These older reports are less standardized and harder to extract consistently, with less frequent observations and fewer comparable analytical fields.

Collecting these filings, extracting the relevant information, and checking that the fields remain comparable across the full date range would be a substantial and time-consuming part of the project. Even after that work, the public filings alone would not provide observed monthly holdings throughout the sample.

We wanted to flag these limitations as we continue assessing the data availability. In parallel, potential sources for the other types of data needed for the project, including fund returns, flows, and benchmark information is still an open question.

Best,
Daksh (*on behalf of the team*)
[Quoted text hidden]

Daksh Gupta <dakshg@andrew.cmu.edu> Wed, Sep 30, 2026 at 8:00 AM
To: "Song,Qiang [VCM]" <qiang_song@vcm.vanguard.com>, "Deshpande,Amit [VCM]" <amit_deshpande2@vcm.vanguard.com>
Cc: Anna Huang <annahua2@andrew.cmu.edu>, Caroline Liu <cmliu@andrew.cmu.edu>, Ziyu He <ziyuhe@andrew.cmu.edu>, Zola Zhang <zolaz@andrew.cmu.edu>, Chad Schafer <cschafer@cmu.edu>

Hi Qiang and Amit,

Please find attached a sample dataset for one fund, combining publicly available SEC N-PORT data with Bloomberg data. We would appreciate your feedback on whether these fields provide a useful basis for the project and how we should refine the scope given the data available.

The sample includes:

- **SEC N-PORT:** Fund identifiers, reporting and filing dates, total assets and liabilities, and security-level holdings, including CUSIP/ISIN where available, reported values, portfolio weights, asset classifications, bond maturities, and coupons. It also includes fund flows and fund-level interest-rate and credit-spread sensitivities.

- **Bloomberg:** Historical NAV, total returns, benchmark information, and fund asset history. Historical expense ratios remain a gap.

I retrieved the Bloomberg data through the Excel API. We still need to identify a scalable approach to retrieve it across the broader fund universe. This is an initial sample; the full dataset remains to be assembled and validated.

Following our discussion, we propose focusing on 2019 onward, when N-PORT reporting began, given the challenges in assembling comparable holdings data for earlier periods.

Could you review the sample and let us know which fields would be most useful, whether any essential information is missing, and what fund universe and research scope would be feasible with this coverage? Your feedback will help us prioritize the next round of data collection.

Best,
Daksh

[Quoted text hidden]

3 attachments

vanguard_core_bond_fund_holdings.csv
2752K

vanguard_core_bond_fund_metadata.csv
6K

VCOBX_Bloomberg_Download.xlsx
335K

Daksh Gupta <dakshg@andrew.cmu.edu> Wed, Sep 30, 2026 at 12:54 PM
To: "Song,Qiang [VCM]" <qiang_song@vcm.vanguard.com>, "Deshpande,Amit [VCM]" <amit_deshpande2@vcm.vanguard.com>
Cc: Anna Huang <annahua2@andrew.cmu.edu>, Caroline Liu <cmliu@andrew.cmu.edu>, Ziyu He <ziyuhe@andrew.cmu.edu>, Zola Zhang <zolaz@andrew.cmu.edu>, Chad Schafer <cschafer@cmu.edu>

Hi Qiang and Amit,

We noticed that the Bloomberg formulas in the previous attachment were returning errors. Please find attached an updated version with the data pasted as values so it can be viewed without a Bloomberg connection.

Thanks,
Daksh

[Quoted text hidden]

VCOBX_BBG_Data.xlsx
15K

Song,Qiang [VCM] <qiang_song@vcm.vanguard.com> Wed, Sep 30, 2026 at 4:51 PM
To: Daksh Gupta <dakshg@andrew.cmu.edu>, "Deshpande,Amit [VCM]" <amit_deshpande2@vcm.vanguard.com>
Cc: Anna Huang <annahua2@andrew.cmu.edu>, Caroline Liu <cmliu@andrew.cmu.edu>, Ziyu He <ziyuhe@andrew.cmu.edu>, Zola Zhang <zolaz@andrew.cmu.edu>, Chad Schafer <cschafer@cmu.edu>

Hi team,

Thank you for the sample dataset.

Below is guidance on fields, structure, benchmarks and derivatives. The derivatives section is the longest because it is where this project is most likely to go wrong, and it is specific to fixed income in a way that equity fund studies never have to deal with.

1. Fields that matter most

Of the hundred or so fields you extracted, a much smaller set will carry the analysis.

Priority	Fields	Why
Essential	Interest rate sensitivity at each maturity point; credit spread sensitivity, investment grade and below investment grade	The fund's own risk system output. Includes derivatives, correctly netted. This is your primary measure of positioning.
Essential	Monthly returns by share class; monthly flows (sales, redemptions, reinvestment)	Monthly, where holdings are only quarterly. Flows are needed to separate deliberate repositioning from accommodating investor money.
Core	Security identifier, reported value and weight, maturity date, coupon rate and type, asset and issuer category	The holdings table itself.
Core	The entire derivative block	See section 5. Do not discard this.
Keep	Payables for delayed delivery purchases	Reveals mortgage forward activity. In your sample this moved from 0.69bn to 4.02bn in one quarter.
Discard	Most other balance sheet fields	Almost always zero.

On the risk fields: they are reported in dollars. Divide by net assets and multiply by ten thousand and you get the same figures expressed in years, which is how the industry reads them. For your sample fund at March 2026 this gives total duration of **5.70 years** and investment grade spread duration of **3.53 years**. Those two numbers describe most of what the manager is doing.

2. How to organize the data

Please do not flatten this into a single table. The data genuinely arrives at three frequencies, and forcing it together will either duplicate rows or discard information.

Table	Grain	Key	Contents
Fund quarter	One row per fund per quarter	fund_id + report_date	Net assets, risk metrics, benchmark name, filing date, reporting lag
Holdings	One row per security per fund per quarter	fund_id + report_date + security_id	All positions. Keep derivatives here, but flag them.
Fund month	One row per fund per month	fund_id + month_end	Returns and flows, unpacked from the month one, two and three fields

One trap worth flagging. Month one is the *first* month of the fiscal quarter, not the most recent. Getting this backwards would silently corrupt everything downstream.

Two further points. Always carry the filing date alongside the report date, so any analysis can be restricted to information that was public at the time. And when you assemble the full universe, keep every fund that ever filed, including those that later closed or merged. Building the universe from a current list of funds would quietly exclude the poor performers, which are precisely the funds this project is most interested in.

3. Finding the benchmark

Your plan rests on comparing each fund's positions against its benchmark, and you correctly identified that you have benchmark returns but not benchmark composition. There is a way around this that costs nothing.

Index funds file exactly the same reports as active funds. So an index fund tracking a given benchmark can stand in for the benchmark itself, and it arrives in the same format, on the same dates, through the pipeline you have already built.

The filings already tell you which benchmark each fund uses. Your metadata has a designated index field, which for your sample fund reads *Bloomberg U.S. Aggregate Float Adjusted Index*. The procedure is then: read the designated index name for each fund, map it to a suitable index fund, pull that fund's filings through your existing extractor, and join on report date.

Benchmark	Index fund to use	Filer ID	Portfolio ID
Bloomberg U.S. Aggregate **Float Adjusted**	Vanguard Total Bond Market Index Fund	794105	S000002564
Bloomberg U.S. Aggregate (unadjusted)	iShares Core U.S. Aggregate Bond ETF	1100663	S000004362
Bloomberg U.S. Aggregate (alternates)	SPDR Portfolio Aggregate Bond ETF Schwab U.S. Aggregate Bond ETF	1064642 1454889	S000017334 S000033156
Government mandates	Corresponding Treasury index funds	—	—

Match the *stated* benchmark rather than assuming. The float adjusted and unadjusted versions of the Aggregate differ, and using the wrong one introduces small but systematic errors.

Two cautions. Index funds hold a representative sample rather than every constituent, so this works well for aggregate measures such as sector weights and duration, and poorly for individual security weights — do not use it to claim a fund is overweight a specific bond. Also check that report dates align, because not every fund family uses a calendar fiscal year.

4. Calculating active allocation

Once you have both sides the calculation is straightforward, but two details matter.

Normalize first. Reported weights do not sum to one hundred percent. In your sample the fund sums to 102.0 percent and the index fund to 99.7 percent, because of cash, payables and rounding. Divide each weight by its own total before comparing, otherwise the difference in totals contaminates every sector.

Then subtract. Active weight is the fund's normalized weight minus the benchmark's normalized weight, sector by sector. By construction these sum to zero, which is a useful check on your arithmetic.

Running this on your sample fund for March 2026:

Sector	Fund %	Benchmark %	Active (pp)
Treasuries	20.7	48.1	**−27.4**
Corporates	43.2	29.2	**+14.0**
Non-US sovereign	10.4	1.7	**+8.7**
Agency mortgages	18.4	15.7	+2.7
Other agency	7.2	4.9	+2.4
Municipals	0.1	0.4	−0.3

A useful single summary is half the sum of the absolute active weights, which gives the share of the portfolio positioned differently from the benchmark. Here that is **27.8 percent** at sector level.

> **Now the important part, and the reason I have set this out at length.**
>
> Read on the weights alone, this fund appears to be running an enormous interest rate bet: it holds barely half the Treasury exposure of its benchmark. Yet its reported duration is **5.70 years against the benchmark's 5.65 years** — essentially identical.
>
> Both facts are true. The fund has sold Treasuries, bought corporates, and restored the missing interest rate exposure using derivatives. Its real position is a large credit overweight with no interest rate view at all.
>
> If you had analyzed this fund by weights alone, you would have concluded the opposite of what is happening. Please treat this as the governing lesson of the exercise: **in fixed income, sector weights tell you where the money is, and only the risk measures tell you where the risk is.** You need both, and they answer different questions.

5. Derivatives, and why they decide the answer

In your sample fund, derivatives account for **0.1 percent of the portfolio by market value**. In weight terms they are a rounding error. Measured by notional amount, the interest rate swaps alone come to **32.7 billion dollars against net assets of 21.0 billion**, roughly one and a half times the size of the fund.

Both numbers are correct and neither is the risk. A swap has almost no value when it is struck: two parties agree to exchange payments, neither pays the other at the outset, and the position enters the portfolio at close to zero. What it carries is exposure. A fund can move its duration by a full year through swaps without its holdings table looking any different.

Instrument	What it does	Why managers use it
Interest rate swaps *largest category*	One side pays fixed, receives floating (usually the Secured Overnight Financing Rate). Paying fixed **shortens** duration; receiving fixed **lengthens** it.	Adjusting duration this way is faster and cheaper than trading several hundred million dollars of bonds. Your sample holds 751.5m notional paying 3.1574 percent fixed, maturing March 2028.

Treasury futures	Duration exposure at specific curve points; separate contracts for 2, 5, 10 and 30 year maturities.	The usual tool for expressing a view on the *shape* of the curve rather than its level.
Swaptions	Options on swaps. Risk is **not** proportional to notional.	Positioning for changes in interest rate volatility, and hedging large moves. Handle separately; do not add their notional to the swap total.
Currency forwards	Hedge currency risk on bonds issued outside the United States.	Most core funds hedge almost completely, so these usually *reduce* risk. Your fund's 10.4 percent non-US sovereign holding is very likely hedged this way — confirm rather than assume.
Credit default swap indices	Credit exposure in one liquid instrument instead of hundreds of bonds.	Adding or reducing credit risk quickly. Your sample holds two series of the North American investment grade index.
Mortgage forwards (TBAs) *not flagged as derivatives*	Economically forward contracts on mortgage pools, but they appear in your holdings table as **ordinary mortgage securities**. Recognize them by very short settlement dates and descriptions beginning FNCL.	Maintaining mortgage exposure. Funds roll them month after month. **If you measure turnover by comparing holdings between quarters, these rolls appear as enormous trading activity when nothing is being decided.**

My recommendation on handling all this: use the fund's own reported risk figures as the primary measure of positioning. They come from the manager's risk system, already account for every derivative, netted with the correct sign, and are the only complete measure available to you. Building your own from the holdings table would be substantial work to arrive at a worse answer.

Use the derivative records themselves to describe *how* exposure is being taken rather than how much. Whether a fund expresses a duration view through swaps and futures or through cash bonds is an interesting question in its own right, and as far as I know nobody has examined it systematically across bond funds. Your data supports it.

If you do want to build exposures from holdings as a cross check, the rule is that risk is approximately **notional multiplied by the duration of the underlying**, signed by the direction of the position. Notional alone tells you very little, since a two year and a thirty year swap of the same size differ in risk by more than a factor of ten.

6. Suggested next step

Pull the index fund filings for each benchmark in your universe and produce, for every fund and quarter, the **active sector weights alongside the active duration and active spread duration**. That single table is the core measurement of the project, and everything else builds on top of it.

Do come back with questions. Happy to go through any of this on a call if that would be easier than email.

Best regards,
Qiang

Qiang Song
Quantitative Research Strategist, Fixed Income | Senior Specialist
Vanguard
qiang_song@vcm.vanguard.com

VCM Internal Use Only

[Quoted text hidden]

CONFIDENTIALITY STATEMENT. The information contained in this e-mail message, including attachments, is the confidential information of, and/or is the property of, Vanguard. The information is intended for use solely by the individual or entity named in the message. If you are not an intended recipient or you received this in error, then any review, printing, copying, or distribution of any such information is prohibited, and please notify the sender immediately by reply e-mail and then delete this e-mail from your system.

Song,Qiang [VCM] <qiang_song@vcm.vanguard.com> Tue, Oct 6, 2026 at 2:27 PM
To: Daksh Gupta <dakshg@andrew.cmu.edu>, "Deshpande,Amit [VCM]" <amit_deshpande2@vcm.vanguard.com>
Cc: Anna Huang <annahua2@andrew.cmu.edu>, Caroline Liu <cmliu@andrew.cmu.edu>, Ziyu He <ziyuhe@andrew.cmu.edu>, Zola Zhang <zolaz@andrew.cmu.edu>, Chad Schafer <cschafer@cmu.edu>

Hi team,

An update on the data question.

Prof. Chad Schafer has established that a Morningstar subscription would cost around twenty thousand dollars, which is well beyond what is reasonable for a capstone project. That decision is settled. We will build this from free public sources, with Bloomberg used selectively where nothing free will do.

The practical consequence is that holdings will be observed quarterly rather than monthly. Everything else we need is obtainable.

Sample period: 2019 to the present. The first public N-PORT filings cover holdings as of September, October or November 2019, depending on each fund's fiscal year, with monthly returns for the three months up to each of those dates. Every existing fund on the attached list has data from that point. The two exceptions are Vanguard Core-Plus Bond and Vanguard STAR Core-Plus, which launched later and enter the sample when they began filing. Please do not try to extend the sample before 2019 using the older filing formats. It would take a great deal of document parsing for very little research value.

What to collect

Five free sources. The first two you are already using.

Source	Gives you	Where
1. N-PORT quarterly datasets	Holdings, fund risk measures, monthly returns, monthly flows, derivatives	The SEC structured data sets you are already pulling, one file per quarter from 2019Q4
2. N-PORT for index funds	Benchmark composition and benchmark risk measures	Same files. The index funds to use are listed in the attached file
3. Prospectus risk and return summary data sets *new — please add this*	**Expense ratios**, management fees, fee waivers	sec.gov/dera/data/mutual-fund-prospectus-risk-return-summary-data-sets

4. Federal Reserve yield curve	Daily Treasury zero-coupon curve, for interest rate factors	federalreserve.gov, nominal yield curve page
5. FRED	Credit spread series, for credit factors	fred.stlouisfed.org

On source 3. The SEC publishes the fee tables from fund prospectuses as structured quarterly files, and they key on the same fund and share class identifiers you are already using, so the join is exact. The fields you want are ExpensesOverAssets and NetExpensesOverAssets.

Which files from the N-PORT datasets

The quarterly archive contains over thirty tables and nearly two gigabytes. You do not need most of it. These are the ones that matter, in rough order of importance.

File	Contains
INTEREST_RATE_RISK	The interest rate sensitivities, by currency and maturity point. This is the single most important table in the whole archive.
FUND_REPORTED_INFO	Net assets, **the credit spread sensitivities**, monthly flows, delayed delivery payables. Note the credit measures live here rather than with the interest rate ones.
MONTHLY_TOTAL_RETURN	Monthly returns by share class
SUBMISSION	Report date, filing date, filer identity
FUND_REPORTED_HOLDING	The holdings themselves. Nearly a gigabyte per quarter, so filter to your universe before loading.
IDENTIFIERS and DEBT_SECURITY	Security identifiers, maturity dates, coupons
MONTHLY_RETURN_CAT_INSTRUMENT	Monthly profit and loss split by derivative type. Worth keeping; very few people use it.
The derivative tables	Swaps, futures, forwards, swaptions. Pull as needed for the derivative analysis.

What to take from Bloomberg

You have Bloomberg access, and it should be used deliberately. Terminal data limits will become binding quickly at this scale, so the principle is to use it only where nothing free will serve, and to spend the allowance on the items below rather than on data you can obtain elsewhere.

Priority	What to pull	Why

1	**Credit ratings for individual bonds.** Composite rating and the individual agency ratings, pulled as static reference data per security, not as a time series.	The only genuinely irreplaceable item. Nothing free provides it, and it is what allows you to measure credit quality tilts within the portfolio rather than only at fund level.
2	**Benchmark index analytics.** Modified duration, option-adjusted spread, yield, and sector and quality composition for the Aggregate and its Treasury, corporate, mortgage and high yield sub-indices. Monthly, 2019 onward.	Very small volume for very high value. These are the actual index figures rather than a proxy, and the sub-index returns double as factors for your risk model.
3	**Fund reference data.** One row per fund: inception date, category, stated benchmark, fund family.	Gives you the peer group classification the filings lack. Inception date matters more than it appears, for the reason set out below.

On ratings and volume. The union of securities across the full universe will run to several hundred thousand identifiers, which is more than you should attempt in one job. Three things make it manageable. Pull ratings once per security rather than once per fund and quarter, since the same bond appears in many portfolios. Restrict the pull to positions above a weight threshold, or to the largest holdings in each fund, and report the coverage you achieve. And unless you have a specific reason to need point-in-time ratings, take the current rating as an approximation and note the limitation. Coverage of eighty or ninety percent of portfolio weight is ample for the analysis we have in mind.

On inception date. I ran your sample fund's full return history this week. It beat its benchmark by 63 basis points a year over the period, which looks impressive until you separate the years: 2020 alone contributed 2.46 percentage points, and every other year contributed between zero and 0.6. The fund launched shortly before, grew from 1.3 to 4.3 billion dollars through 2020, and was deploying new money into a dislocated market. That is a consequence of when it launched, not evidence of skill, and excluding 2020 reduces its information ratio from 1.30 to 0.83. Any fund in your universe that launched near a market dislocation will show the same flattering pattern, so fund age and flows need to be controls in the analysis rather than afterthoughts.

What not to use Bloomberg for. Fund returns, because the filings are better: Bloomberg drops funds that have closed, so a return series built from it would be biased toward survivors, which is the specific problem this project is meant to avoid. Expense ratios, because source 3 above provides them free. Fund duration, because the figure the fund reports comes from its own risk system and already includes derivatives, whereas Bloomberg's is an estimate. And any per-security time series, which will exhaust your data allowance immediately.

Practical notes. Use static requests for reference data and historical requests only where you genuinely need history, since the latter consume far more of your allowance. Test field names on a handful of securities before launching a large job. Batch in blocks of a few thousand, spread large pulls across several days, and write every result to disk as you go so that a failed run never has to re-query. Finally, check early whether your entitlement covers index analytics, as that is sometimes licensed separately from the core terminal. If it is not included, the index fund approach remains the fallback and nothing is lost.

The fund universe (attached)

We will focus on the main active core and core-plus bond funds: the large ones, and those that compete directly with Vanguard. The attached file lists them, with the SEC series ID and filer ID for each, so it joins directly to your extraction.

Group	Funds	How to use it

Primary	49	The peer group for the analysis: 3 Vanguard funds (Core Bond, Core-Plus Bond, STAR Core-Plus) and 46 competitors, including PIMCO, Dodge & Cox, Capital Group, Baird, JPMorgan, PGIM, Fidelity, TCW, DoubleLine, BlackRock and others.
Adjacent	4	Related but not core: a corporate-heavy fund, two short-intermediate funds and a multisector fund. Collect them, but keep them out of the core comparison.
Check	2	Vanguard's active ETFs, which are near copies of the Core and Core-Plus mutual funds. Use them as a cross-check only, so the same strategy is not counted twice.
Benchmark	4	Index funds tracking the Bloomberg U.S. Aggregate, used to stand in for the benchmark.

Every identifier has been checked against the 2026 first quarter filings. Every primary fund has also been confirmed as active using the index-fund flag in the funds' Form N-CEN filings. When you extend the list, use that flag rather than fund names to tell index funds from active ones. Several index funds do not have "index" in their names.

Treat the list as a starting point, not the complete universe. It is a snapshot of today's large funds, and building the panel only from today's funds would exclude the ones that closed or shrank, which are often the poor performers this project most wants to study. Western Asset is the clearest example. Its Core Plus fund held 41.6 billion dollars at the end of 2021 and 3.7 billion today; its Core Bond fund fell from 20.9 billion to 1.8 billion. Both are on the list, but a size screen applied to today's assets would have removed them. So for each quarter, check the filings for other core and core-plus funds that were large *at that time*, and apply any size screen using assets at the start of each period rather than today.

Benchmarks

Use the Bloomberg U.S. Aggregate as the common benchmark for every primary fund. Most of them show it in their prospectus as the required broad-based index. More importantly, we are comparing funds against one another, so measuring each one against the same yardstick keeps the comparison about positioning rather than about which benchmark each manager happened to choose.

Please also record each fund's own stated benchmark from its prospectus or fact sheet, in the empty column in the file. It is about an hour's work for 49 funds. Known exceptions are Vanguard Core Bond, which uses the Float Adjusted Aggregate (nearly identical), the two Vanguard core-plus funds, which use the Universal, and Bridge Builder Core Plus, which uses a custom blend.

One warning. The "designated index" field in N-PORT, which you used in the sample, is **not** the performance benchmark. It is a reference portfolio for the SEC's derivatives risk rule. Several funds fill it with their own portfolio or "N/A", and some name an index that is plainly not their benchmark. I have kept it in the file for reference only.

Align dates carefully. Funds report holdings at the end of their own fiscal quarters, not calendar quarters. JPMorgan, Fidelity and T. Rowe Price, for example, report as of November, February, May and August, while Vanguard Core Bond reports as of calendar quarter-ends. Two consequences follow. First, when you compare funds at a given point in time, use each fund's most recent snapshot available as of that date, and record how old it is. Second, compare each fund with its benchmark *as of the same month-end* as the fund's own report. Bloomberg's monthly index analytics make this straightforward, because they are available for every month. The index fund filings are not, since those funds also report on their own fiscal calendars. Join everything on series ID rather than fund name, because names change; Allspring Core Plus, for instance, appears in 2019 simply as "Core Plus Bond Fund".

Two data cautions

> **Reported credit spread duration is not comparable across fund families.** The four index funds in the file all track the same index. Their reported durations agree closely, between 5.70 and 5.88 years, but their reported investment grade spread durations range from **1.90 to 3.20 years**. The likely cause is that filers differ on whether mortgage securities count as credit risk. So interest rate duration can be compared across any funds, but spread duration should only be compared within a fund family, or after you have worked out and adjusted for each family's convention. Please check this before ranking funds on credit risk.
>
> **Screen for implausible values.** Neuberger Berman Core Bond reports 10.2 years of duration, which is not plausible for a core fund. Another filer reported a figure in the thousands. Set sensible bounds, flag what falls outside them, and report how many observations you exclude.

Finally, compute each fund's tracking error against the Aggregate from monthly returns. Some funds that are legally active, such as the rules-based DFA portfolio, may sit very close to the index, and tracking error is the objective way to flag them.

Order of work

I would suggest this sequence.

First, assemble the fund quarter table with risk measures, returns, flows and expense ratios for the funds on the list. This is small, fast, and sufficient for a surprising amount of analysis on its own.

Second, pull the benchmark analytics and fund reference data from Bloomberg. Both are small jobs, and running them early confirms what your entitlement covers before anyone invests effort in something larger.

Third, add the benchmark comparison using the index fund filings, and extend the universe back through time as described above.

Fourth, bring in the holdings, which are far bulkier and can wait until the first three are working. Leave the ratings pull until the universe is final, so that you request the security list once rather than three times.

If you can produce active duration, active credit spread duration and active sector weights for every fund and quarter, you will have the core measurement of the project and everything else builds on top of it.

Do come back with questions. Happy to take any of this on a call.

Best regards,
Qiang

VCM Internal Use Only

From: Daksh Gupta <dakshg@andrew.cmu.edu>
Sent: Wednesday, September 30, 2026 12:55 PM
To: Song,Qiang [VCM] <qiang_song@vcm.vanguard.com>; Deshpande,Amit [VCM] <amit_deshpande2@vcm.vanguard.com>
Cc: Anna Huang <annahua2@andrew.cmu.edu>; Caroline Liu <cmliu@andrew.cmu.edu>; Ziyu He <ziyuhe@andrew.cmu.edu>; Zola Zhang <zolaz@andrew.cmu.edu>; Chad Schafer <cschafer@cmu.edu>
Subject: [External] Re: Weekly Update - Vanguard / CMU Capstone Project

[Quoted text hidden]

[Quoted text hidden]

capstone_fund_universe.csv
8K

Daksh Gupta <dakshg@andrew.cmu.edu> Wed, Oct 7, 2026 at 11:30 AM
To: qiang_song@vcm.vanguard.com, amit_deshpande2@vcm.vanguard.com
Cc: annahua2@andrew.cmu.edu, cmliu@andrew.cmu.edu, ziyuhe@andrew.cmu.edu, zolaz@andrew.cmu.edu, cschafer@cmu.edu

Hi Qiang and Amit,

Thank you for the detailed emails. The guidance is very helpful and gives us a clear starting point for data collection.

We were not able to spend much time on the project last week because of multiple project and presentation deadlines in our other classes. We also have finals this week, so we have been balancing the capstone with our other coursework.

Best,
Daksh

Song,Qiang [VCM] <qiang_song@vcm.vanguard.com> Wed, Oct 7, 2026 at 3:27 PM
To: Daksh Gupta <dakshg@andrew.cmu.edu>, "Deshpande,Amit [VCM]" <amit_deshpande2@vcm.vanguard.com>
Cc: "annahua2@andrew.cmu.edu" <annahua2@andrew.cmu.edu>, "cmliu@andrew.cmu.edu" <cmliu@andrew.cmu.edu>, "ziyuhe@andrew.cmu.edu" <ziyuhe@andrew.cmu.edu>, "zolaz@andrew.cmu.edu" <zolaz@andrew.cmu.edu>, "cschafer@cmu.edu" <cschafer@cmu.edu>

Hi team,

Since you're in the middle of midterms, let's skip this week's meeting and give you the time back.

Please keep working on the data, and reach out if any questions come up.

Best,

Qiang

VCM Internal Use Only

From: Daksh Gupta <dakshg@andrew.cmu.edu>
Sent: Wednesday, October 7, 2026 11:31 AM
To: Song,Qiang [VCM] <qiang_song@vcm.vanguard.com>; Deshpande,Amit [VCM] <amit_deshpande2@vcm.vanguard.com>
Cc: annahua2@andrew.cmu.edu; cmliu@andrew.cmu.edu; ziyuhe@andrew.cmu.edu; zolaz@andrew.cmu.edu; cschafer@cmu.edu
Subject: [External] Re: Weekly Update - Vanguard / CMU Capstone Project

[Quoted text hidden]

[Quoted text hidden]

Song,Qiang [VCM] <qiang_song@vcm.vanguard.com> Fri, Oct 23, 2026 at 11:02 AM
To: Anna Huang <annahua2@andrew.cmu.edu>, "dakshg@andrew.cmu.edu" <dakshg@andrew.cmu.edu>, "ziyuhe@andrew.cmu.edu" <ziyuhe@andrew.cmu.edu>, "zolaz@andrew.cmu.edu" <zolaz@andrew.cmu.edu>, "cmliu@andrew.cmu.edu" <cmliu@andrew.cmu.edu>
Cc: "Deshpande,Amit [VCM]" <amit_deshpande2@vcm.vanguard.com>, "cschafer@cmu.edu" <cschafer@cmu.edu>

Hi team,

Thank you for the sample run. All six steps work and the results match ours, which means the pipeline is reading the filings correctly. Good work, especially while juggling exams.

Below are three corrections from the sample run, and then the next step: organizing each fund's positioning into two layers.

Three corrections from the sample run

1. Measure swap direction by risk, not notional. Your slide shows $26.8bn of pay-fixed swaps against $5.9bn of receive-fixed, which suggests the swaps shorten duration. Measured by risk, they do the opposite. Most of the pay-fixed swaps are short-dated and the receive-fixed swaps are long-dated, so on our rough estimate the swaps **add** about 0.6 years of duration overall. For each swap, estimate risk as notional times the duration of the swap (driven by its remaining maturity), with a sign based on direction. This is the "notional tells you little" point, in your own data.

2. Separate interest rate swaps from other swaps. We count 56 interest rate swaps (27 pay-fixed, 29 receive-fixed), not 65. The other nine include two credit default swap index positions, which are credit trades and belong under credit risk, not interest rates.

3. Compute benchmark weights yourselves from the index fund filing, matched to the fund's report date, rather than using the figures from my email.

Next step: positioning in two layers

So far we have looked at sector weights and fund-level risk numbers. The aim now is to describe each fund the way a portfolio manager would: which decisions did the manager make relative to the benchmark? Doing that cleanly needs two layers, because they answer different questions.

	Layer 1: Asset classification	Layer 2: Factor exposures
Question	Where is the money?	Where is the risk?
Unit	Each holding goes into exactly one sector	Each holding can contribute to several factors at once
Measured in	Percent of net assets, normalized to 100%	Years of duration or spread duration, or percent weight, depending on the factor
Derivatives	Kept as a separate line; their market value is close to zero	Fully included; this is where most of their effect shows up

A ten-year corporate bond is a good example. In Layer 1 it sits in one place, investment grade corporates. In Layer 2 it adds to US duration, to the long end of the curve, to credit risk and to the credit quality profile, all at once.

Both layers are measured **relative to the benchmark**, every quarter, for every fund.

Layer 1: Asset classification

Assign every holding to one of the sectors below. The weights should add to 100% for the fund, and again for the benchmark. The active weight is the difference.

Sector	How to identify it in N-PORT
US Treasury (nominal)	Debt, issuer type Treasury, not inflation-linked
TIPS	Issuer type Treasury, inflation-linked (security description, or CUSIP checked against the Treasury's TIPS list)
Agency and other government	Issuer type government agency or government-sponsored enterprise, excluding mortgage securities
Agency MBS	Mortgage asset category with an agency issuer. Flag TBAs separately (very short settlement, descriptions starting FNCL, FGLMC, G2SF and similar)
Non-agency MBS and CMBS	Mortgage asset category with a corporate issuer. Telling residential from commercial needs the security description
ABS	Asset-backed categories other than mortgages, including CLOs
Investment grade corporate	Corporate debt rated BBB- or above. **Needs ratings from Bloomberg**
High yield corporate	Corporate debt rated below BBB-. **Needs ratings**
Non-US developed sovereign	Issuer type non-US sovereign, developed-market country
Emerging markets	Sovereign or corporate issuer in an emerging-market country. Define the country list once and use it throughout
Municipal	Issuer type municipal
Cash and equivalents	Short-term investments, repo, money market funds, plus reported cash
Derivatives	Kept as a separate line at market value, which will be close to zero. Their risk is handled in Layer 2

Use the asset category and issuer type codes first, and fall back to the security description only where the codes are not enough. Record which rule placed each holding, so the classification can be checked. Until you have ratings, keep investment grade and high yield corporates together as one temporary "corporate" sector.

Layer 2: Factor exposures

These eight factors describe the main decisions a core or core-plus manager makes. For each, measure the fund's position minus the benchmark's.

Factor	What it measures	How to compute it
1. US duration	Long or short interest rate risk	Sum of the reported USD rate sensitivities at the five maturity points, converted to years, minus the benchmark's
2. US yield curve	Positioned for steepening or flattening	Active exposure at each maturity point. A simple summary is active long-end exposure (10 and 30 years) minus active short-end exposure (1 and 5 years). This is where the swap and futures positions show up most clearly
3. Non-US rates and FX	Interest rate and currency exposure outside the US	Rate sensitivity for each non-USD currency (reported where above 1% of assets). For currency: non-USD holdings by currency, plus or minus currency forwards, to give net unhedged exposure
4. MBS	Over- or underweight mortgages	Agency MBS active weight from Layer 1, including TBAs. Report non-agency separately
5. TIPS, breakevens and swap spreads	Inflation and swap-versus-Treasury positioning	TIPS active weight from Layer 1, plus inflation swaps (derivatives referencing CPI). For swap spreads, compare interest rate swap risk with Treasury risk. Treat this one as approximate
6. Credit risk	How much credit spread risk the fund takes	Reported investment grade and high yield spread sensitivities, converted to years, minus the benchmark's. Add the CDX positions here. Remember the earlier warning: compare reported spread duration across fund families with care
7. Credit quality	Up or down in quality	Active weight by rating bucket (AAA, AA, A, BBB, high yield) and the average rating, versus the benchmark. **Needs Bloomberg ratings**
8. Rates and credit hedging	Whether the rate position offsets the credit position in a sell-off	Not computed from holdings. Regress monthly fund returns, over a rolling window, on a rate factor and a credit factor, and look at the two sensitivities together, and how they move with each other

For the sample fund, we would expect roughly the following: neutral on US duration, overweight on credit risk, a curve position driven largely by derivatives, and a large Treasury underweight that does not show up as a duration bet. Running both layers on VCOBX first is a good way to check the code before scaling up.

Data needed

Source	What to take	Used for

N-PORT, holdings FUND_REPORTED_HOLDING, IDENTIFIERS, DEBT_SECURITY	Asset category, issuer type, issuer country, currency, value and weight, maturity, coupon type, security description	Layer 1 sectors; factors 3, 4 and 5
N-PORT, risk measures INTEREST_RATE_RISK, FUND_REPORTED_INFO	Rate sensitivity by currency and maturity point; investment grade and high yield spread sensitivity; net assets	Factors 1, 2, 3 and 6
N-PORT, derivatives swap, futures, forward and swaption tables	Notional, direction, maturity, reference index or instrument, currency	How each factor is expressed; checking the reported figures; CDX for factor 6; inflation swaps for factor 5
N-PORT, monthly returns	Monthly returns for one reference share class per fund	Factor 8
Index fund filings	The same fields, for the benchmark proxies in the fund list	Benchmark side of every comparison
Bloomberg, security level	Composite credit rating for each bond; optionally a sector code, for example to identify financials	Investment grade versus high yield in Layer 1; factor 7
Bloomberg, index level	For the Aggregate, monthly: duration at each maturity point, spread duration, and sector and rating weights. Also the Treasury, corporate, MBS and high yield sub-index returns	A cleaner benchmark for both layers; factor returns for factor 8
Federal Reserve and FRED	Treasury yield curve; investment grade and high yield credit spread series	Rate and credit factors for factor 8, and checking reported duration

The target output is one row per fund per quarter, with the active weight for each Layer 1 sector and the active position on each of the eight factors. That table is the core measurement of the project. Comparing it between funds that persistently outperform and those that persistently lag is the analysis.

Suggested order: Layer 1 and factors 1 to 4 and 6 first, since they need only N-PORT and the index fund filings. Then add ratings for factor 7, and the return regression for factor 8.

Best regards,
Qiang

Confidential - IT System Information

From: Anna Huang <annahua2@andrew.cmu.edu>
Sent: Wednesday, October 21, 2026 3:37 PM
To: Song,Qiang [VCM] <qiang_song@vcm.vanguard.com>
Subject: [External] Re: Weekly Update - Vanguard / CMU Capstone Project

Hello Qiang,

Attached please find the powerpoint for this week.

Best regards,

Anna

Confidential - IT System Information

[Quoted text hidden]